

Impact of electrostatic correlations on the frequency response of a graphene electrode in ionic liquid

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We investigate the frequency response of the graphene-based electrochemical systems containing room-temperature ionic liquids using the method of matched asymptotic expansion in the limit of thin double-layers to study the diffuse-charge dynamics of graphene electrodes interfaced with the ionic liquids. The theory in the framework of the asymptotic matching is mainly supported by (1) the length scale $\sqrt{\lambda_D L_c}$ that suitably characterizes the structure of the double layer in ionic liquids, with λ_D referring to the Debye length and L_c referring to the electrostatic correlation length, and (2) the charging time, $\tau_c = \lambda_D^{3/2} L / (D L_c^{1/2})$, where D is the diffusion coefficient and L is the thickness of the electrolyte. With this response time, the diffuse-charge dynamics is deduced at the leading order to obtain an analytic expression for low-frequency impedance. As a result of the new length scaling $\sqrt{\lambda_D L_c}$, we additionally study the behavior of double-layer in graphene-metal electrochemical systems involving ionic liquids, where electrostatic correlation length L_c is found to play a prominent role in the linear regime. The results are then compared with the conventional electrochemical cells containing a metal-metal electrode pair.

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I. INTRODUCTION

Benefited from a large ionic density and a broad window of electrochemical stability [1], ionic liquids (ILs) are distinguished from ordinary diluted electrolytes in various applications, including ionic conductive membranes in fuel cells [2,3], solvents in biological media [4], capacitors, and supercapacitors [5–7]. However, optimization of such applications is still a considerable challenge, primarily attributed to the confounding dynamics of charge distributions at the electrode/IL interface [8]. For particular applications, such as batteries and supercapacitors, in which a high capacitance is desirable, the issue has shown to be mitigated by interfacing ILs with electrochemically inert and easily processable carbon-based electrodes [9], such as activated carbon [10], carbon nanotubes [11], and graphene [12]. In experiments, among the available approaches to observe the charge-diffuse dynamics in ILs [13], electrochemical impedance spectroscopy (EIS) proves to be an effective technique, offering a robust performance to charge distributions at the graphene/IL interface [14,15]. Thus, a deeper analysis of the frequency response associated with the graphene electrodes interfaced with ILs holds extreme relevance.

To model the electrode/IL interface in the presence of constant charge or potential on the electrode (static regime), mean-field theories are often applied to predict the charge distribution in the electric-double layer (EDL) near the surfaces. According to the related theories, constitutions of ions at the interface of electrode/IL reveal the existence of two distinct regimes: overscreening, attributed to the electrostatic correlation effects, and crowding, associated with steric effects [16]. We note that modeling of the electrode/IL interfaces in static regime is mainly dedicated to the semi-infinite domain in the literature [16–19], with only a few papers studying the finite domain [20,21].

On the other hand, in the dynamic regime, e.g., in the presence of a time-dependent externally applied potential [22,23], ILs can only be studied on a finite domain. Motivated by the raised issues in theories and applications, here we revisit our recent study of impedance spectroscopy in simple electrolyte solutions in the presence of graphene electrode [24,25] by generalizing it to ILs using mean-field theories adopted from Bazant, Storey, and Kornyshev (BSK) [16]. We use the boundary conditions derived in [26], where electrostatic correlation effects are nonzero at the electrode surfaces in combination with a modified boundary condition associated with the graphene [24,25] interfaced with the ILs. The BSK model has been utilized in electrolytic solutions, effectively capturing a range of ion correlation phenomena,

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such as the reversal of electrophoretic mobility in colloidal particles within multivalent electrolytes [27] and the facilitation of ion transport through biological ion channels [28–31]. Contrary to our previous work, the DC analysis is omitted in this study and the system is assumed to be in the neutral equilibrium state, allowing us to mainly focus on the AC response of the system in ILs. Following the procedures outlined by Zhao [22], the boundary-layer analysis of the modified Poisson-Nernst-Planck (mPNP) equations in ILs is performed at the leading order in the limit of thin EDLs, with a suitable length scale characterizing the structure of EDLs, as well as a proper time scale adjusting the asymptotic matching.

In this paper, we begin by linearizing the mPNP equations containing the electrostatic correlation effects and steric effects under the condition that the surface potentials are small compared to the thermal voltage in conjunction with the boundary conditions involving the nonzero correlation effects at the electrode surfaces and blocking nature of electrodes accompanied by an updated boundary condition related to the graphene electrode in the presence of ILs [32]. Having normalized the equations, asymptotic matching is performed at the leading order in the frequency domain using the new length scales for the thin EDLs and response time in ILs. With the results derived in the outer region and the dynamics of the inner region at the leading order, an analytic expression for the impedance at low frequency is obtained. The discussion in Sec. IV “Results and Discussion” is then completed by studying the structure of EDL and the frequency response of graphene-metal (GM) and metal-metal (MM) electrode configurations in ILs with the “Conclusions and Future Directions” presented in Sec. V.

II. THEORY

We consider an electrochemical cell along the X axis, containing a symmetric monovalent binary IL with identical bulk concentrations c_b at equilibrium. The two ion species have equal diffusion coefficients $D_+ = D_- = D$ and mobilities $\mu_+ = \mu_- = \mu$. The IL is confined between a graphene electrode (GE) at $X = -L$ and a metal electrode (ME) at $X = L$, which are symmetrically positioned around the origin, $X = 0$. An externally time-dependent voltage, $V(\tau)$, is applied to the system which leads to the distribution of an electrostatic potential, $\Phi(X, \tau)$, across the cell accompanied by the potential drop inside the doped graphene, $V_q(\tau)$, and the potential drop between the two electrodes in the region occupied by the IL as $V_{IL}(\tau) = \Phi(L, \tau) - \Phi(-L, \tau)$. Therefore, the total potential difference applied to the ME with respect to the GE is $V(\tau) = V_q(\tau) + V_{IL}(\tau)$.

Later, through asymptotic matching, we will demonstrate that $V_{IL}(\tau)$ can be decomposed into a linear potential drop across the “outer region” (the bulk of the IL) and the “inner regions” (the diffuse layers) adjacent to the electrodes. Additionally, we will show that the total current in the outer region is approximated by the conduction current, whereas in the inner regions, it is primarily governed by the displacement current at each electrode.

In a one-dimensional system, the transport of charged species with the concentrations $C_{\pm}(X, \tau)$ and fluxes $F_{\pm}(X, \tau)$

are connected through

$$\frac{\partial C_{\pm}(X, \tau)}{\partial \tau} = -\frac{\partial F_{\pm}(X, \tau)}{\partial X}, \quad (1)$$

where $F_{\pm}(X, \tau)$ is obtained through the modified Nernst-Planck equation, taking into account the steric effects [22]

$$F_{\pm}(X, \tau) = -D \left[\frac{\partial C_{\pm}(X, \tau)}{\partial X} \pm \frac{e}{k_B T} \frac{\partial \Phi(X, \tau)}{\partial X} C_{\pm}(X, \tau) + \frac{\gamma C_{\pm}(X, \tau)}{1 - \gamma [C_+(X, \tau) + C_-(X, \tau)]} \times \frac{\partial}{\partial X} [C_+(X, \tau) + C_-(X, \tau)] \right], \quad (2)$$

where we used the Einstein’s relation, $\mu/D = e/k_B T$. In the above, T is the temperature, k_B is the Boltzmann constant, and $e > 0$ is the proton charge. In Eq. (2), γ is the volume available to each ion within a lattice-gas model [17], so that the product $\gamma [C_+(X, \tau) + C_-(X, \tau)]$ then represents the total volume fraction occupied by ions at a given point. (We ignore possible ion pairing in the bulk [33,34].) This formulation accounts for steric (finite-size) effects by enforcing a crowding limit, since the local ion concentration cannot exceed the maximum packing limit where $\gamma [C_+(X, \tau) + C_-(X, \tau)] = 1$.

In Eq. (2), we assumed that, in the mean-field approximation, the ion flux is driven by the electric field given by $\mathbf{E} = -\nabla \Phi$. This potential in the BSK model [16] is determined by the modified Poisson equation assuming that the displacement field is given by $\mathbf{D} = \hat{\epsilon} \mathbf{E}$, where the permittivity $\hat{\epsilon}$ is an integral operator taking into account short-range electrostatic correlations. In the one-dimensional case, the Poisson equation reads

$$-\hat{\epsilon} \frac{\partial^2 \Phi(X, \tau)}{\partial X^2} = \rho(X, \tau) = e[C_+(X, \tau) - C_-(X, \tau)], \quad (3)$$

where $\rho(X, \tau)$ is the total ionic charge density and the effective permittivity is given by [16]

$$\hat{\epsilon} = \epsilon_b \left(1 - L_c^2 \frac{\partial^2}{\partial X^2} \right), \quad (4)$$

with ϵ_b being the bulk permittivity and L_c the short-range electrostatic correlation length. Physically, L_c captures the nonlocal interactions arising from ion-ion correlations that go beyond the mean-field description of classical Poisson theory. When $L_c = 0$, from Eq. (4), the standard Poisson equation is recovered, implying no short-range correlations. A finite L_c introduces higher-order spatial derivatives, allowing the model to account for effects, such as overscreening and charge oscillations observed in concentrated electrolytes and ILs [16,35]. Substituting Eq. (4) into Eq. (3) gives the modified fourth-order Poisson equation

$$\epsilon_b \left(L_c^2 \frac{\partial^2}{\partial X^2} - 1 \right) \frac{\partial^2 \Phi(X, \tau)}{\partial X^2} = e[C_+(X, \tau) - C_-(X, \tau)], \quad (5)$$

involving the impacts of the electrostatic correlation length L_c on the potential distribution $\Phi(X, \tau)$.

For applied potentials much smaller than the thermal voltage, $V(\tau) \ll k_B T/e$, the equations can be linearized by the

fact that the sum of ion concentrations in the linear regime is constant, i.e., $C_+(X, \tau) + C_-(X, \tau) \approx 2c_b$. Therefore, the ion flux, Eq. (2), is linearized as

$$F_{\pm}(X, \tau) \approx -D \left[\frac{\partial C_{\pm}(X, \tau)}{\partial X} \pm \frac{e}{k_B T} \frac{\partial \Phi(X, \tau)}{\partial X} C_{\pm}(X, \tau) \right], \quad (6)$$

indicating that the steric effects, represented by γ , are not included in the linear regime.

By subtracting Eq. (1), the continuity equation is derived for the total charge density,

$$\frac{\partial \varrho(X, \tau)}{\partial \tau} = -\frac{\partial J_c(X, \tau)}{\partial X}, \quad (7)$$

where $J_c(X, \tau) = e[F_+(X, \tau) - F_-(X, \tau)]$, is the linearized ionic conduction current that can be expressed in terms of the electrostatic potential $\Phi(X, \tau)$ by substituting Eq. (5) into Eq. (6) and subtracting to obtain

$$J_c(X, \tau) = \epsilon_b D \left[\frac{\partial^3 \Phi(X, \tau)}{\partial X^3} - \frac{1}{\lambda_D^2} \frac{\partial \Phi(X, \tau)}{\partial X} - L_c^2 \frac{\partial^5 \Phi(X, \tau)}{\partial X^5} \right], \quad (8)$$

where $\lambda_D = \sqrt{\epsilon_b k_B T / 2e^2 c_b}$ is the Debye screening length.

In the presence of ILs, the displacement field is modified as $D(X, \tau) = -\epsilon \partial \Phi(X, \tau) / \partial X$, so that the displacement current, $J_d(X, \tau) = -\partial D(X, \tau) / \partial \tau$, taking into account Eq. (4), is modified according to

$$J_d(X, \tau) = -\epsilon_b \frac{\partial}{\partial \tau} \left[\frac{\partial \Phi(X, \tau)}{\partial X} - L_c^2 \frac{\partial^3 \Phi(X, \tau)}{\partial X^3} \right], \quad (9)$$

which is written as the contribution of a classically defined displacement field (first term), and the IL contribution (second term) where electrostatic correlation among the species is present.

The explicit solution to the above set of equations requires the appropriate boundary conditions. As demonstrated by de Souza and Bazant [26], the appropriate boundary constraint for the higher-order Poisson equation is based on an interfacial stress balance, which involves the mechanical and electrochemical forces from the structured ionic layers near the charged interface. Specifically, by enforcing the continuity of the normal component of the total stress tensor (including both hydrostatic and electrostatic contributions), one derives a natural condition on the electrostatic potential, such that $\partial^3 \Phi(X, \tau) / \partial X^3 \neq 0$, at electrode surfaces, which is related to the jump in the normal component of the Maxwell stress tensor across the electrode-electrolyte boundary. Physically, this nonvanishing third derivative encodes the electrostatic stress exerted by the correlated ionic atmosphere on the surface, and is directly tied to the short-range repulsion and overscreening effects characteristic of ionic liquids. This refined boundary condition addresses a key limitation in the original BSK theory [16], where such interfacial stresses were neglected in which case the boundary condition reads $\partial^3 \Phi(X, \tau) / \partial X^3 = 0$. Therefore, to improve the mathematical consistency of the model and enhance the physical realism in capturing the

layered structure and oscillatory screening observed in experiments and molecular simulations of IL systems, we take the non-negligible electrostatic correlations at the electrode surfaces [26]

$$L_c \frac{\partial^3 \Phi(X, \tau)}{\partial X^3} \Big|_{X=\pm L} = \frac{\partial^2 \Phi(X, \tau)}{\partial X^2} \Big|_{X=\pm L}, \quad (10)$$

and vanishing conduction current at the blocking electrode surfaces

$$\left[\frac{\partial^3 \Phi(X, \tau)}{\partial X^3} - \frac{1}{\lambda_D^2} \frac{\partial \Phi(X, \tau)}{\partial X} - L_c^2 \frac{\partial^5 \Phi(X, \tau)}{\partial X^5} \right] \Big|_{X=\pm L} = 0. \quad (11)$$

We note that the nonvanishing third derivative in the boundary condition Eq. (10) is consistent with the linear regime in this work, where the surface potentials are considered small, i.e., $V(\tau) \ll k_B T / e$ [26]. For large surface potentials, $V(\tau) \gg k_B T / e$, where the system drives the transport processes out of the equilibrium state thermodynamically, it appears that a boundary condition proposed by Gupta *et al.* [36] is more appropriate, giving $\partial^2 \Phi(X, \tau) / \partial X^2 = 0$.

In the following, as shown in Eqs. (11) and (12) of our previous work [24] for two different electrode materials (graphene and metal), one can use the zero-flux conditions at the electrodes and invoke charge neutrality in the Gauss's law applied to the entire IL cell to show that the system remains antisymmetric, i.e., $\Phi(-X, \tau) = -\Phi(X, \tau)$, $\varrho(-X, \tau) = -\varrho(X, \tau)$, and $C_{\pm}(-X, \tau) = C_{\pm}(X, \tau)$, allowing us to consider only half of the cell.

To include the interfacial properties of graphene, we introduce the total diffuse surface charge density (per unit area) near the GE as [37]

$$\varsigma_d(\tau) = \int_{-L}^0 \varrho(X, \tau) dX. \quad (12)$$

The linearized surface charge density of graphene, $\varsigma_g(\tau)$, is given by [24]

$$\varsigma_g(\tau) \approx -C_q V_q(\tau), \quad (13)$$

where $C_q = 4e^2 k_B T \ln 2 / \pi (\hbar u_F)^2 \approx 0.8 \mu\text{F}/\text{cm}^2$ is the quantum capacitance of neutral graphene at room temperature with $\hbar$ being the reduced Planck's constant and $u_F \approx 10^6$ m/sec being the Fermi speed associated with the π bands. Moreover, $V_q(\tau)$ is obtained from the instantaneous potential value by employing the Dirac cone approximation for the π -electronic bands of graphene [24].

Next, we introduce the dimensionless parameters following Ref. [22,24]. We take L as the reference length scale and $\tau_c = \lambda_D^{3/2} L / (D L_c^{1/2})$ as the reference time scale. Time and space are then represented by $t = \tau / \tau_c$ and $x = X / L$, and the problem is reformulated through reduced variables: $l_c = L_c / L$ for the correlation length, $l_D = \lambda_D / L$ for the Debye length, $\rho = \varrho / (2ec_b)$ for the charge density, $\{\phi, v\} = (e/k_B T)\{\Phi, V\}$ for the potentials, and $j = J / J_{\text{lim}}$ for the current densities (per unit area) in which $J_{\text{lim}} = 2eDc_b / L$ is the diffusion-limited current density. The surface charge densities (per unit area) are then normalized according to $\sigma = \varsigma / (2ec_b L)$ with any capacitance in the system normalized by the Debye capacitance (per unit area), $C_D = \epsilon_b / \lambda_D$. The equations are then transformed

into the frequency domain by substituting $\partial/\partial t \equiv s = i\Omega\tau_c$ using the Laplace transform where Ω (rad/sec) is the frequency of the AC applied voltage, $V(\tau)$.

As a result of such normalizations and transformation, two additional length scales appear in the equations. To introduce them, take the continuity equation, Eq. (7), for instance, when normalized and transformed,

$$s\rho(x, s) = -\frac{\epsilon}{\delta_c} \frac{\partial j_c(x, s)}{\partial x}. \quad (14)$$

In this equation, $\delta_c = l_c/l_D = L_c/\lambda_D$ is recognized as the characteristic correlation length that controls the extent of the overcompensation of electrode surface charge by the diffuse layer. Overcompensation occurs, e.g., at a negatively charged electrode, when the diffuse layer attains more positive charges than what is needed to neutralize the surface charge of the electrode. To balance and restore the local charge to maintain electroneutrality, the negative charges that were initially repelled return from the bulk to screen the excessive positive charge by creating a spatial charge oscillation within the EDL as positive and negative ions alternatively accumulate and balance each other in successive layers, a natural consequence of overscreening in ILs. Speaking of charge oscillations, their characteristic wavelength can be determined by the second length scale, $\epsilon = \sqrt{\lambda_D L_c}/L$, which characterizes the structure of the EDL by controlling its thickness, which is in contrast to the commonly used Debye screening length scale λ_D in aqueous solutions. We will show in the next section that this length scale forms the basis of asymptotic matching used in this study.

Returning to normalizing and transforming the equations, the conduction current, Eq. (8), is normalized according to

$$j_c(x, s) = \frac{\epsilon^4}{l_c^2} \frac{\partial^3 \phi(x, s)}{\partial x^3} - \frac{\partial \phi(x, s)}{\partial x} - \epsilon^4 \frac{\partial^5 \phi(x, s)}{\partial x^5}, \quad (15)$$

with the displacement current, Eq. (9), being normalized and transformed as

$$j_d(x, s) = -\epsilon s \left[\frac{\partial \phi(x, s)}{\partial x} - l_c^2 \frac{\partial^3 \phi(x, s)}{\partial x^3} \right], \quad (16)$$

along with the Poisson equation, Eq. (5),

$$\rho(x, s) = \epsilon^4 \frac{\partial^4 \phi(x, s)}{\partial x^4} - \frac{\epsilon^4}{l_c^2} \frac{\partial^2 \phi(x, s)}{\partial x^2}. \quad (17)$$

The boundary conditions, Eq. (10) and Eq. (11), are also normalized according to

$$\delta_c \frac{\partial^3 \phi(x, s)}{\partial x^3} \Big|_{x=\pm 1} = \frac{\partial^2 \phi(x, s)}{\partial x^2} \Big|_{x=\pm 1}, \quad (18)$$

$$\left[\frac{\epsilon^4}{l_c^2} \frac{\partial^3 \phi(x, s)}{\partial x^3} - \frac{\partial \phi(x, s)}{\partial x} - \epsilon^4 \frac{\partial^5 \phi(x, s)}{\partial x^5} \right] \Big|_{x=\pm 1} = 0. \quad (19)$$

The total surface charge density of the diffuse layer near GE, Eq. (12), is expressed in normalized form as

$$\sigma_d(s) = \int_{-1}^0 \rho(x, s) dx, \quad (20)$$

and the surface charge density of graphene, Eq. (13), is given in normalized form as

$$\sigma_g(s) \approx \frac{-\epsilon}{\sqrt{\delta_c}} C_q v_q(s), \quad (21)$$

where $C_q = C_q/C_D$ is the normalized quantum capacitance of neutral graphene at room temperature. We should stress here that the appearance of $\sqrt{\delta_c}$ in the expression for surface charge density of graphene, Eq. (21), is a required condition to perform the asymptotic analysis explained in the next section and is a direct consequence of adopting the length scale $\sqrt{\lambda_D L_c}$ in ILs.

III. ASYMPTOTIC MATCHING

In this study, the Debye length and the electrostatic correlation length are $\lambda_D \sim 1 \text{ \AA}$ and $L_c \sim 1 \text{ nm}$, with the size of the electrochemical cell being $L \sim 1 \text{ \mu m}$. From this, we can conclude that $\epsilon = \sqrt{\lambda_D L_c}/L \ll 1$, which suggests that the EDL is *thin*, enabling us to perform asymptotic expansions around ϵ and apply the method of matched asymptotic expansion in the singular limit of thin EDLs in ILs where $\epsilon \ll 1$ holds. In the following, we outline the asymptotic matching presented in [24,37] and focus on leading order dynamics of the diffuse layer near the GE and finally derive an expression for the impedance of the cell.

We adopt the method of matched asymptotic expansions to analytically resolve the multiscale nature of the EDL and bulk charge dynamics in a finite domain. Through a suitable length-scale analysis, this method divides the entire thickness of the IL cell into two inner regions and an outer region, a hallmark of the boundary-layer theory. While the inner region analysis effectively treats the two interfaces with electrodes as if they were extended into semi-infinite ILs, it is crucial in a dynamic (as opposed to static) regime to maintain current conservation throughout the IL cell. This necessarily includes the electro-neutral outer region of finite thickness, which supplies finite resistance to the overall impedance of the cell.

A. Outer region

We start by considering a regular asymptotic expansion, indicated by a bar accent, in the bulk “outer” region as

$$\bar{\phi}(x, s) \sim \bar{\phi}_0(x, s) + \epsilon \bar{\phi}_1(x, s) + \epsilon^2 \bar{\phi}_2(x, s) + \dots \quad (22)$$

and similar expansions for $\bar{\rho}(x, s)$, $\bar{j}_c(x, s)$, $\bar{j}_d(x, s)$, etc. Substituting such expansions into the scaled equations of the preceding subsection, i.e., Eqs. (14)–(21), it follows from Eq. (17) that $\bar{\rho}_0(x, s) = 0$. This verifies the electroneutrality of the outer region at the leading order in ϵ , namely $\bar{\rho}(x, s) \approx \bar{\rho}_0(x, s) = 0$. The conduction current, Eq. (15), at the leading order is

$$\bar{j}_{c,0}(x, s) = -\frac{\partial \bar{\phi}_0(x, s)}{\partial x}, \quad (23)$$

with the displacement current, Eq. (16), vanishing at the leading order, $\bar{j}_{d,0}(x, s) = 0$. This suggests that the leading order total current, $\bar{j}_0(s) = \bar{j}_{c,0}(x, s) + \bar{j}_{d,0}(x, s)$, involving

Eq. (23), is

$$\bar{j}_0(s) = -\frac{\partial \bar{\phi}_0(x, s)}{\partial x}, \quad (24)$$

in the outer region which is the contribution of only the conduction current at the leading order. By integrating Eq. (24), the antisymmetric leading order potential profile in the outer region is

$$\bar{\phi}_0(x, s) = -\bar{j}_0(s)x, \quad (25)$$

as it should owing to a direct consequence of proper choice for the time scale τ_c . Therefore, the total leading order potential drop in the outer region is

$$\bar{v}_{b,0}(s) = \bar{\phi}_0(1, s) - \bar{\phi}_0(-1, s) = -2\bar{j}_0(s). \quad (26)$$

B. Inner region

The regular outer approximations need to be matched with singular “inner” approximations within the boundary layers. Due to the symmetry of the problem about the origin, where $\phi(-x, s) = -\phi(x, s)$, we focus only on the boundary layer at the GE located at $x = -1$. To analyze this region, we introduce the inner coordinate transformation $y = (x + 1)/\epsilon$. This rescaling eliminates singularities in the equations, allowing us to perform a regular asymptotic expansion for the inner approximation, which is represented using tilde accents.

$$\tilde{\psi}(y, s) \sim \tilde{\psi}_0(y, s) + \epsilon \tilde{\psi}_1(y, s) + \epsilon^2 \tilde{\psi}_2(y, s) + \dots, \quad (27)$$

where $\tilde{\psi}(y, s) = \tilde{\phi}(y, s) - \bar{\phi}(-1, s)$ is the excess voltage relative to the outer region. Therefore, the continuity equation, Eq. (14), is transformed into the inner region as

$$s\tilde{\rho}(y, s) = -\frac{1}{\delta_c} \frac{\partial \tilde{j}_c(y, s)}{\partial y}. \quad (28)$$

The conduction current, Eq. (15), in the inner region is transformed as

$$\epsilon \tilde{j}_c(y, s) = \frac{1}{\delta_c} \frac{\partial^3 \tilde{\psi}(y, s)}{\partial y^3} - \frac{\partial \tilde{\psi}(y, s)}{\partial y} - \frac{\partial^5 \tilde{\psi}(y, s)}{\partial y^5}, \quad (29)$$

with the displacement current, Eq. (16), being transformed in the inner region as

$$\tilde{j}_d(y, s) = -s \left[\frac{\partial \tilde{\psi}(y, s)}{\partial y} - \delta_c \frac{\partial^3 \tilde{\psi}(y, s)}{\partial y^3} \right]. \quad (30)$$

The Poisson equation, Eq. (17), in the inner region is expressed

$$\tilde{\rho}(y, s) = \frac{\partial^4 \tilde{\psi}(y, s)}{\partial y^4} - \frac{1}{\delta_c} \frac{\partial^2 \tilde{\psi}(y, s)}{\partial y^2}. \quad (31)$$

As for the boundary conditions, Eq. (18) and Eq. (19), they take the following form in the inner region:

$$\left. \frac{\partial^3 \tilde{\psi}(y, s)}{\partial y^3} \right|_{y=0} = \frac{\epsilon}{\delta_c} \left. \frac{\partial^2 \tilde{\psi}(y, s)}{\partial y^2} \right|_{y=0}, \quad (32)$$

$$\left[\frac{1}{\delta_c} \frac{\partial^3 \tilde{\psi}(y, s)}{\partial y^3} - \frac{\partial \tilde{\psi}(y, s)}{\partial y} - \frac{\partial^5 \tilde{\psi}(y, s)}{\partial y^5} \right] \Big|_{y=0} = 0, \quad (33)$$

in addition to the fact that the potential decays to zero far away from the electrode surfaces in the inner region

$$\tilde{\psi}(\infty, s) = 0. \quad (34)$$

The diffuse surface charge density near the GE, Eq. (20), has the scaling $\sigma_d(s) \sim \epsilon \tilde{\sigma}_d(s)$, where

$$\tilde{\sigma}_d(s) = \int_0^\infty \tilde{\rho}(y, s) dy \sim \tilde{\sigma}_{d,0}(s) + \epsilon \tilde{\sigma}_{d,1}(s) + \epsilon^2 \tilde{\sigma}_{d,2}(s) + \dots, \quad (35)$$

and the surface charge density of graphene, Eq. (21), has the scaling $\sigma_g(s) = \epsilon \tilde{\sigma}_g(s)$, where

$$\tilde{\sigma}_g(s) \approx \frac{-1}{\sqrt{\delta_c}} C_q \tilde{v}_q(s) \sim \tilde{\sigma}_{g,0}(s) + \epsilon \tilde{\sigma}_{g,1}(s) + \epsilon^2 \tilde{\sigma}_{g,2}(s) + \dots. \quad (36)$$

C. Leading order dynamics

To analyze the linear dynamics in the inner region, we first combine Eqs. (28), (29), and (31), to obtain a single governing equation in terms of the potential $\tilde{\psi}(y, s)$,

$$\frac{1}{\delta_c} \frac{\partial^6 \tilde{\psi}(y, s)}{\partial y^6} - \left(\frac{1}{\delta_c^2} + \epsilon s \right) \frac{\partial^4 \tilde{\psi}(y, s)}{\partial y^4} + \frac{1}{\delta_c} (1 + \epsilon s) \frac{\partial^2 \tilde{\psi}(y, s)}{\partial y^2} = 0. \quad (37)$$

At the leading order ($\epsilon = 0$), the above reduces to

$$\frac{\partial^6 \tilde{\psi}_0(y, s)}{\partial y^6} - \frac{1}{\delta_c} \frac{\partial^4 \tilde{\psi}_0(y, s)}{\partial y^4} + \frac{\partial^2 \tilde{\psi}_0(y, s)}{\partial y^2} = 0, \quad (38)$$

whose bounded solution satisfying Eq. (34) at leading order, $\tilde{\psi}_0(\infty, s) = 0$, is given by

$$\tilde{\psi}_0(y, s) = A_1(s) \frac{e^{-\gamma_1 y}}{\gamma_1^2} + A_2(s) \frac{e^{-\gamma_2 y}}{\gamma_2^2}, \quad (39)$$

where

$$\gamma_1 = \sqrt{\frac{1 + \sqrt{1 - 4\delta_c^2}}{2\delta_c}}, \quad \gamma_2 = \sqrt{\frac{1 - \sqrt{1 - 4\delta_c^2}}{2\delta_c}}. \quad (40)$$

To determine integration constants $A_1(s)$ and $A_2(s)$, we require that the boundary conditions, Eqs. (32) and (33), at the leading order are

$$\left. \frac{\partial^3 \tilde{\psi}_0(y, s)}{\partial y^3} \right|_{y=0} = 0, \quad (41)$$

$$\left[\frac{1}{\delta_c} \frac{\partial^3 \tilde{\psi}_0(y, s)}{\partial y^3} - \frac{\partial \tilde{\psi}_0(y, s)}{\partial y} - \frac{\partial^5 \tilde{\psi}_0(y, s)}{\partial y^5} \right] \Big|_{y=0} = 0. \quad (42)$$

The above two boundary conditions can be combined into a single boundary condition when Eq. (41) is plugged into Eq. (42), giving

$$\left[\frac{\partial \tilde{\psi}_0(y, s)}{\partial y} + \frac{\partial^5 \tilde{\psi}_0(y, s)}{\partial y^5} \right] \Big|_{y=0} = 0. \quad (43)$$

It is interesting to note that, referring to Eqs. (32) and (41), the boundary condition that involves correlation effects at the electrode, developed in Ref. [26], is reduced to the boundary condition that neglects those effects, which was proposed in the original form of the BSK theory [16], when we only consider the leading order theory of the inner layer.

The additional boundary condition is derived by integrating the Poisson equation, Eq. (31) from 0 to ∞ and invoking the usual van Dyke matching conditions for the electric field and its higher orders,

$$\begin{aligned} \lim_{y \rightarrow \infty} \frac{\partial \tilde{\psi}(y, s)}{\partial y} &\sim \epsilon \lim_{x \rightarrow -1} \frac{\partial \bar{\phi}(x, s)}{\partial x} \implies \frac{\partial \tilde{\psi}_0(y, s)}{\partial y} \Big|_{y=\infty} = 0, \\ \lim_{y \rightarrow \infty} \frac{\partial^3 \tilde{\psi}(y, s)}{\partial y^3} &\sim \epsilon^3 \lim_{x \rightarrow -1} \frac{\partial^3 \bar{\phi}(x, s)}{\partial x^3} \implies \frac{\partial^3 \tilde{\psi}_0(y, s)}{\partial y^3} \Big|_{y=\infty} = 0, \end{aligned} \quad (44)$$

taking into account Eq. (41), to obtain the leading order surface charge density in the diffuse layer

$$\tilde{\sigma}_{d,0}(s) = \frac{1}{\delta_c} \frac{\partial \tilde{\psi}_0(y, s)}{\partial y} \Big|_{y=0}. \quad (45)$$

Also, the surface charge density of graphene, Eq. (36), at the leading order reads

$$\tilde{\sigma}_{g,0}(s) \approx \frac{-1}{\sqrt{\delta_c}} C_q \tilde{v}_{q,0}(s), \quad (46)$$

where $\tilde{v}_{q,0}(s)$ is the leading order potential drop inside graphene. This can be related to the total leading order potential drop across the half-cell adjacent to the GE, $\tilde{v}_{g,0}(s) = \tilde{v}_{q,0}(s) + \tilde{v}_{d,0}(s)$. Here, $\tilde{v}_d(s) = \bar{\phi}(-1, s) - \bar{\phi}(0, s) = -\tilde{\psi}(0, s) \sim -\tilde{\psi}_0(0, s) - \epsilon \tilde{\psi}_1(0, s) - \dots$ is the potential drop across the diffuse layer near the GE so that Eq. (46) is

$$\tilde{\sigma}_{g,0}(s) \approx \frac{-1}{\sqrt{\delta_c}} C_q [\tilde{v}_{g,0}(s) + \tilde{\psi}_0(0, s)]. \quad (47)$$

The surface charge densities, Eqs. (45) and (47), can be combined to enforce charge neutrality at leading order near GE, $\tilde{\sigma}_{g,0}(s) + \tilde{\sigma}_{d,0}(s) = 0$, giving rise to the following boundary condition for graphene [25], which reads at the leading order:

$$\tilde{\psi}_0(0, s) - \frac{1}{\sqrt{\delta_c} C_q} \frac{\partial \tilde{\psi}_0(y, s)}{\partial y} \Big|_{y=0} = -\tilde{v}_{g,0}(s). \quad (48)$$

We mark here that the leading order charge neutrality, $\tilde{\sigma}_{g,0}(s) + \tilde{\sigma}_{d,0}(s) = 0$, used here, is consistent with the generalized Gauss's law in ILs mentioned by Schlumpberger and Bazant (see Eq. (10) in Ref. [38]).

With the pair of boundary conditions derived in Eqs. (43) and (48), the integration constants $A_1(s)$ and $A_2(s)$ are readily obtainable and, when inserted into Eq. (39), yield the leading order potential drop in the inner region adjacent to the

graphene electrode

$$\begin{aligned} \tilde{\psi}_0(y, s) = -\tilde{v}_{g,0}(s) &\left[\frac{e^{-y\gamma_2} \gamma_1 (1 + \gamma_1^4)}{\frac{1}{\sqrt{\delta_c} C_q} (\gamma_1^4 - \gamma_2^4) + \gamma_1 - \gamma_2 + \gamma_1^5 - \gamma_2^5} \right. \\ &\left. - \frac{e^{-y\gamma_1} \gamma_2 (1 + \gamma_2^4)}{\frac{1}{\sqrt{\delta_c} C_q} (\gamma_1^4 - \gamma_2^4) + \gamma_1 - \gamma_2 + \gamma_1^5 - \gamma_2^5} \right]. \end{aligned} \quad (49)$$

We remark that, to the leading order, the dynamic effects in the above potential are only contained in the amplitude factor $\tilde{v}_{g,0}(s)$, whereas its spatial dependence is governed by the factor in square brackets that depends on the inner coordinate $y \in [0, \infty)$ but not on s .

With Eq. (49), we are able to construct the full solutions of the electric potential at leading order as the uniformly valid approximations in space by adding the outer approximation, Eq. (25), and inner approximation, Eq. (49), and subtracting the overlaps. Taking advantages of the symmetry, we obtain the following leading order approximation for the electrostatic potential:

$$\phi(x, s) \sim \bar{j}_0(s)x + \tilde{\psi}_0\left(\frac{1+x}{\epsilon}, s\right) - \tilde{\psi}_0\left(\frac{1-x}{\epsilon}, s\right). \quad (50)$$

D. Conservation of current

It may be verified that the choice of the charging time [22], $\tau_c = \lambda_D^{3/2} L / (D L_c^{1/2})$, is appropriate for ILs by substituting the Poisson equation, Eq. (31), into the continuity equation, Eq. (28), integrating from 0 to ∞ and matching the fields using Eq. (44), together with Eq. (41), to conclude the following leading order relation:

$$-s \frac{\partial \tilde{\psi}_0(y, s)}{\partial y} \Big|_{y=0} = \tilde{j}_{c,0}(\infty, s) - \tilde{j}_{c,0}(0, s). \quad (51)$$

From matching the conduction current at leading order, i.e., $\lim_{y \rightarrow \infty} \tilde{j}_{c,0}(y, s) \sim \lim_{x \rightarrow -1} \bar{j}_{c,0}(x, s)$, considering the blocking electrode boundary condition, $\tilde{j}_{c,0}(0, s) = 0$ from Eq. (43), the right-hand side of Eq. (51) is

$$-s \frac{\partial \tilde{\psi}_0(y, s)}{\partial y} \Big|_{y=0} = \bar{j}_{c,0}(-1, s). \quad (52)$$

By inspecting Eqs. (30) and (41), one can identify the left-hand side of the Eq. (52) as the leading order displacement current,

$$\tilde{j}_{d,0}(0, s) = -s \frac{\partial \tilde{\psi}_0(y, s)}{\partial y} \Big|_{y=0}, \quad (53)$$

so that when substituted into Eq. (52) this leads to

$$\tilde{j}_{d,0}(0, s) = \bar{j}_{c,0}(-1, s). \quad (54)$$

The physical meaning of Eq. (54) corroborates the fact that the current is conserved at the leading order and that the current is continuous across the diffuse layer. This implies that the boundary layer serves as a capacitor whose charging process is controlled by the leading order displacement current, $\tilde{j}_{d,0}(0, s)$, which is fed and supported by the leading order

conduction current, $\bar{j}_{c,0}(-1, s)$, from the bulk, supplying the necessary charges for the displacement current to develop at the electrodes.

E. Impedance

The dimensionless impedance of the cell, normalized to the DC bulk resistance (per unit area), $\mathcal{R}_b = L\lambda_D^2/(\epsilon_b D)$, can be represented as the leading order contributions of the outer and inner regions, given by

$$Z(s) = \frac{-v(s)}{j(s)} = \frac{-\bar{v}_{b,0}(s)}{\bar{j}_{c,0}(-1, s)} - \frac{\tilde{v}_{q,0}(s) + 2\tilde{v}_{d,0}(s)}{\tilde{j}_{d,0}(0, s)}, \quad (55)$$

where $v(s) = \bar{v}_{b,0}(s) + \tilde{v}_{g,0}(s) + \tilde{v}_{d,0}(s)$. From the inner region analysis, the total leading order potential drop across the half-cell adjacent to the GE is given by $\tilde{v}_{g,0}(s) = \tilde{v}_{q,0}(s) + \tilde{v}_{d,0}(s)$. On the other hand, from the outer region analysis [Eq. (26)], the leading order potential drop across the bulk is $\bar{v}_{b,0}(s) = \bar{\phi}_0(1, s) - \bar{\phi}_0(-1, s) = -2\bar{j}_0(s)$. In Eq. (55), $j(s)$ is the total current density, which is in the first term given by $j(s) = \bar{j}_{c,0}(-1, s) = \bar{j}_0(s)$, i.e., the leading order conduction current in the bulk using the result obtained from Eq. (25), and in the second term it is given by $j(s) = \tilde{j}_{d,0}(0, s)$, i.e., the leading order displacement current on the GE surface from Eq. (53). Also, from Eq. (55), we observe that the asymptotic matching models the proposed system as an RC circuit in series connection,

$$Z(s) = R + \tilde{Z}(s), \quad (56)$$

where the first term,

$$R = \frac{-\bar{v}_b(s)}{\bar{j}_{c,0}(-1, s)} = 2, \quad (57)$$

is the resistive term assigned to the outer region, and the second term,

$$\tilde{Z}(s) = -\frac{\tilde{v}_{q,0}(s) + 2\tilde{v}_{d,0}(s)}{\tilde{j}_{d,0}(0, s)} \equiv \frac{1}{s\tilde{C}}, \quad (58)$$

is the capacitive term resulting from the inner region analysis, with $\tilde{C}$ being the total dimensionless capacitance. Recalling $\delta_c = L_c/\lambda_D$, the charging time is rewritten $\tau_c = \delta_c^{-1/2} L\lambda_D/D$, so that the frequency variable, $s = i\Omega\tau_c$, is scaled according to $s = i\omega\delta_c^{-1/2}$ where $\omega = \Omega L\lambda_D/D$ is the scaling for the dimensional frequency Ω that is independent of the correlation parameter δ_c . As a result, the impedance, Eq. (56), may be rewritten as

$$Z(\omega, \delta_c) = 2 + \frac{1}{i\omega C(\delta_c)}, \quad (59)$$

where

$$C(\delta_c) = (C_q^{-1} + 2C_d^{-1}(\delta_c))^{-1} \quad (60)$$

is the total capacitance scaled by C_D and $C_d = C_d/C_D$ is the normalized capacitance across the diffuse layer given by [35]

$$C_d(\delta_c) = \frac{2\sqrt{2}\chi}{(1+\chi)^{3/2} - (1-\chi)^{3/2}}, \quad (61)$$

where $\chi = \sqrt{1 - 4\delta_c^2}$.

IV. RESULTS AND DISCUSSION

To be consistent with the available experimental and simulation data, the physical parameters of the present IL are $T = 300$ K for the temperature, $c_M = 0.5$ M ($M \equiv \text{mol/liter}$) for the ion concentration, and $\epsilon_b = 5\epsilon_0$ for the dielectric permittivity with $\epsilon_0 \approx 8.85 \times 10^{-12}$ F/m being the permittivity of the free space. Therefore, the proper definition of the Debye length gives $\lambda_D \approx 1$ Å and, thus, the Debye capacitance of $C_D \approx 40 \mu\text{F}/\text{cm}^2$, which we find to be consistent with the values reported in Ref. [16].

At low voltages in a linear regime, the solutions have only one free parameter, the normalized correlation length scale δ_c . To capture the effects of the ion-ion correlations across the weak, intermediate, and strong regimes, and to be consistent with the existing results in the literature, we choose the range from $\delta_c = 0$ to $\delta_c = 16$. To study the structure of EDL affected by the correlation length, we first renormalize the leading order potential drop in the diffuse layer, Eq. (49), according to $\tilde{\Psi}_0(y) = -\tilde{\psi}_0(y, s)/\tilde{v}_{g,0}(s)$. We note that the thus-defined normalized potential $\tilde{\Psi}_0(y)$ is independent from s . Next, to expose the impacts of the correlation length, recalling $\epsilon = \sqrt{\lambda_D L_c}/L = \sqrt{\delta_c} l_D$, we rewrite the inner coordinate $y = (1+x)/\epsilon = \zeta/\sqrt{\delta_c}$, where $\zeta = (1+x)/l_D = (L+X)/\lambda_D \in [0, \infty)$ is introduced as the new scaling for the inner coordinate y , which is independent of the parameter δ_c and corresponds to the scaling used in Ref. [35]. This allows us to observe the effects of the correlation length on the diffuse layer potential drop with respect to the rescaled inner coordinate ζ by making a substitution $y = \zeta/\sqrt{\delta_c}$ in $\tilde{\Psi}_0(y)$, for both electrode pairs, as shown in Figs. 1(a) and 1(b) [39].

Based on Eq. (40), we observe two distinctive behaviors for both electrode pairs at the critical value of $\delta_c = 0.5$. In the region $\delta_c < 0.5$, and in particular in the limiting case, $\delta_c \rightarrow 0$, the effects of the IL are weak and both electrode configurations show a smooth and monotonic behavior as depicted in Figs. 1(a) and 1(b). In this weak regime, from Eq. (49), noting that $C_q = C_q/C_D \approx 0.02$, the potential drop of the GM system approaches $\tilde{\Psi}_0(y, s) \rightarrow C_q/(1+C_q) \approx 0.02$ as $\delta_c \rightarrow 0$ which is observable in Fig. 1(a). For the MM system which is achieved by taking the limit $C_q \rightarrow \infty$ [25], we obtain $\tilde{\Psi}_0(y, s) \rightarrow 1$ for all δ_c values and is also observable in Fig. 1(b). For the values falling in the range $\delta_c > 0.5$, the impact of electrostatic correlations is strong and is seen in both electrode pairs. In the limit $\delta_c \rightarrow \infty$, Eq. (49) asymptotes to $\tilde{\Psi}_0(y, s) \rightarrow e^{-y/\sqrt{2}}[\cos(y/\sqrt{2}) - \sin(y/\sqrt{2})]$, highlighting the oscillatory behavior. This also implies that as δ_c increases, the extent of the EDL scales proportionally to $\sqrt{\delta_c}$ since $y \propto \delta_c^{-1/2}$, which is noticeable in Figs. 1(a) and 1(b) for both systems. We should mention here that the observation explained here regarding Fig. 1(b) for the MM system is consistent with the numerical results reported by Storey and Bazant (see panel (a) of Fig. 1 in Ref. [35]), as well as the results obtained by Gupta et al. (see panel (a) of Fig. 12 in Ref. [40] and Fig. 3 in Ref. [41]).

For the GM electrode pair in Fig. 1(a), we observe that the magnitude of the potential drop is significantly smaller compared to the MM electrode pair, Fig. 1(b). This may indicate that graphene reduces the impact of correlations due to its unique electronic properties (i.e., quantum capacitance),

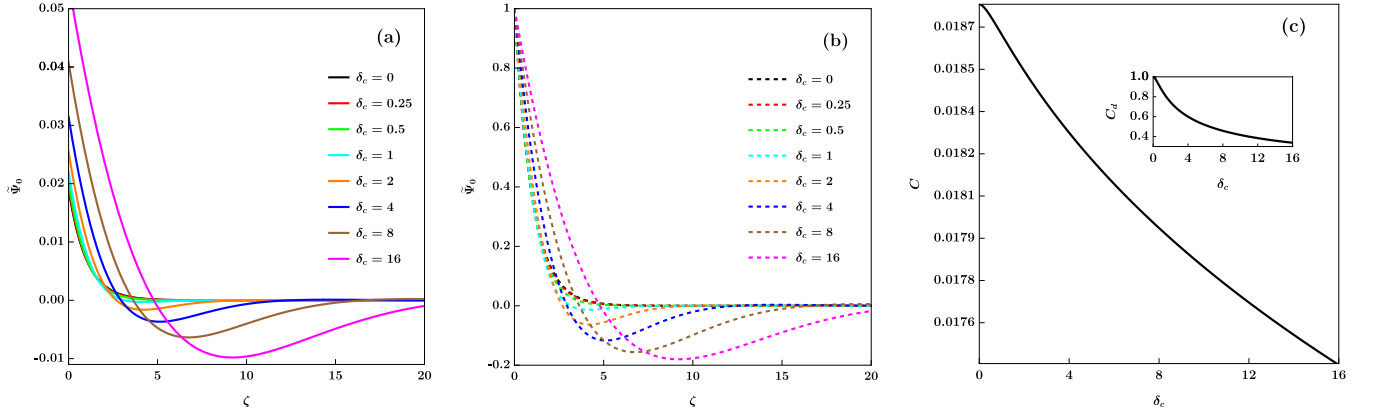


FIG. 1. Impact of the electrostatic correlation length δ_c on the structure of the EDL. (a) Variations of the normalized leading order inner potential drop $\tilde{\Psi}_0$ as a function of the rescaled inner coordinate $\zeta = (L + X)/\lambda_D$ for GM electrode pair. (b) Variations of the normalized leading order inner potential drop $\tilde{\Psi}_0$ as a function of the rescaled inner coordinate $\zeta = (L + X)/\lambda_D$ for MM electrode pair. (c) Variations of the total scaled capacitance $C(\delta_c)$ and the scaled EDL capacitance $C_d(\delta_c)$ (inset) as a function of δ_c .

allowing for enhanced charge distribution uniformity, efficient screening, and significance of ionic transport.

The total capacitance of the GM system with respect to variations of δ_c is also shown in Fig. 1(c). Generally, in a highly correlated IL, the capacitive impacts are negligible and the correlation effects reduce the capacitance [35]. This can be confirmed by the results derived in Eqs. (60) and (61) when $\delta_c \rightarrow \infty$, giving $C \rightarrow 0$ and $C_d \rightarrow 0$, and is observable in Fig. 1(c) for the GM electrode pair and its inset for the MM electrode pair. In the case of weak correlation effects for which we take the limit $\delta_c \rightarrow 0$, the total capacitance of the GM system, Eq. (60), approaches $C \rightarrow C_q/(1 + C_q) \approx 0.02$, and for the MM system, Eq. (61), where $C_q \rightarrow \infty$ gives $C_d \rightarrow 1$, which is in agreement with the numerical results reported by Storey and Bazant (see panel (b) of Fig. 1 in Ref. [35]). Although the total capacitance shows small values, $C \approx 0.02$, for low values of δ_c , this may indicate the observability of graphene electrodes in ILs in the low-frequency measurements, a similar conclusion as was drawn in our recent work in the case of aqueous electrolytes [25].

Figure 2 illustrates the frequency response of the system influenced by the correlation length δ_c . The response is characterized by the phase, $\phi = \arctan(Z_i/Z_r)$ shown in Fig. 2(a) and the magnitude, $|Z| = \sqrt{Z_r^2 + Z_i^2}$, shown in Fig. 2(b) where Z_r and Z_i represent the real and imaginary components of the scaled impedance Z in Eq. (59) with respect to the scaled frequency $\omega = \Omega L \lambda_D / D$ shown in a log scale. For a typical room-temperature IL with a diffusion coefficient of $D = 10^{-7}$ cm²/sec and the electrolyte thickness $L = 1 \mu\text{m}$, the charging time factor is estimated as $L \lambda_D / D \sim 11 \mu\text{sec}$, indicating that the charging processes in ILs take longer compared to aqueous electrolytes where we reported 3 μsec in our recent study [24].

According to the phase response, for the MM configuration, the phase transitions occur over a broader range of frequencies compared to the GM configuration for all values of δ_c . At high frequencies with a reduction in δ_c , both configurations asymptotically show resistive behavior when $\phi \rightarrow 0^\circ$. At low frequencies, MM exhibits stronger

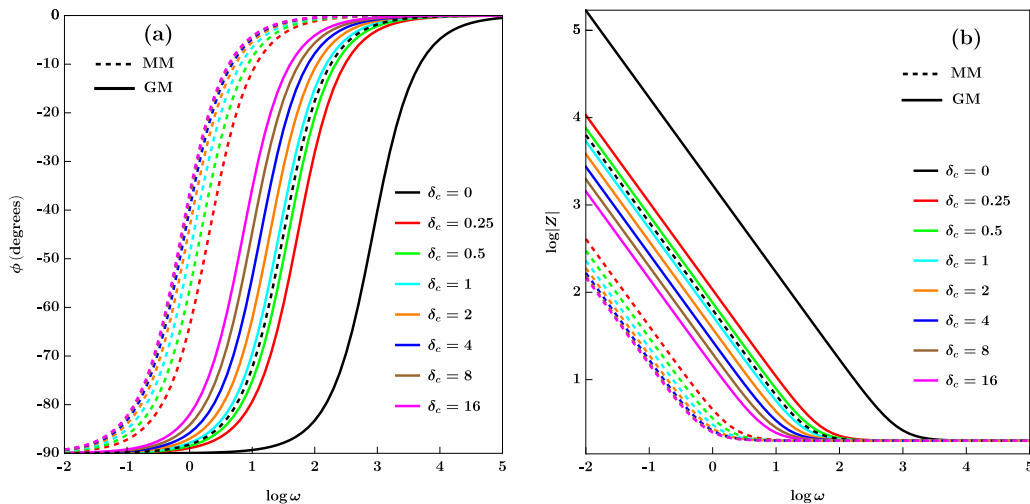


FIG. 2. Impact of the electrostatic correlation length δ_c on the frequency response. (a) Variations of the phase response for different values of δ_c . (b) Variations of the magnitude response for different values of δ_c .

capacitive behavior than GM as δ_c increases, but GM seems to be more tunable in terms of low frequency owing to exhibiting capacitive behavior, $\phi = -90^\circ$, at slightly higher frequencies. As for the magnitude response, in the low-frequency regime, GM configuration shows a higher magnitude of impedance compared to MM for the same δ_c with a slope of nearly -1 , indicating a higher capacitive effect at low frequencies. Both MM and GM configurations converge to similar $\log |Z|$ values at high frequencies with the zero slope, suggesting the same resistive response at high frequencies for all δ_c . The MM configuration consistently exhibits a lower $\log |Z|$ than GM across the mid-frequency range, highlighting the improved conductive properties of MM.

V. CONCLUSIONS AND FUTURE DIRECTIONS

In this study, we have analytically derived the impedance expression for an electrochemical cell containing the room-temperature IL in a neutral equilibrium state that is flanked by a graphene and a metallic electrode pair under an AC modulation of the external voltage. The method of matched asymptotic expansion is performed to a cell of finite thickness $2L$ in the limit of thin double layers, $\epsilon = \sqrt{\lambda_D L_c}/L \ll 1$, to analyze the leading order diffuse-charge dynamics using the charging time scale, $\tau_c = \lambda_D^{3/2} L/(D L_c^{1/2})$, proposed by Zhao [22].

In a linear regime under the influence of the electrostatic correlation length parameter, $\delta_c = L_c/\lambda_D$, the structure of the EDL and the frequency response are investigated. From the numerical results for the potential drop over the diffuse layer and the total capacitance, we find results that agree with the relevant works [22,35] for the metal-metal electrode configuration. In the case of a graphene-metal (GM) electrode pair, we observe a lower potential drop in the diffuse layer compared to the metal-metal (MM) systems, which may

be relevant to the related applications regarding low energy storage capacity, minimized ohmic losses, high-rate electrochemical reactions, and enhanced stability of IL cells. Also, our results show that the presence of the graphene electrode in ILs reduces the capacitance significantly compared to traditional MM systems, which may be relevant for low-frequency measurements such as biosensing, as mentioned in our recent work [25]. Our results for the frequency response under the impacts of the correlation length reveal that MM configurations are capable of covering larger frequency spectrum and displaying stronger low-frequency capacitive behavior. However, the low-frequency capacitive behavior is arguably more tunable in the GM systems.

As a future work, we will add the DC analysis to the present study to observe the frequency response of the GM systems in an equilibrium state with charged graphene and reveal the effects of not only the electrostatic correlation length, but also the finite size of ions (steric effects) in ILs and capture the interplay between the dynamics of the overscreening and the crowding under the influence of high DC voltages.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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- [1] R. L. Vekariya, A review of ionic liquids: Applications towards catalytic organic transformations, *J. Mol. Liq.* **227**, 44 (2017).
 - [2] U. A. Rana, M. Forsyth, D. R. MacFarlane, and J. M. Pringle, Toward protic ionic liquid and organic ionic plastic crystal electrolytes for fuel cells, *Electrochim. Acta* **84**, 213 (2012).
 - [3] H. Nakamoto and M. Watanabe, Brønsted acid–base ionic liquids for fuel cell electrolytes, *Chem. Commun.* 2539 (2007).
 - [4] C. A. Summers and R. A. Flowers, Protein renaturation by the liquid organic salt ethylammonium nitrate, *Protein Sci.* **9**, 2001 (2000).
 - [5] A. Scalia, A. Varzi, A. Moretti, P. Ruschhaupt, A. Lamberti, E. Tresso, and S. Passerini, Electrolytes based on n-butyl-n-methyl-pyrrolidinium 4, 5-dicyano-2-(trifluoromethyl) imidazole for high voltage electrochemical double layer capacitors, *ChemElectroChem* **6**, 552 (2019).
 - [6] A. Scalia, A. Varzi, A. Lamberti, E. Tresso, S. Jeong, T. Jacob, and S. Passerini, High energy and high voltage integrated photo-electrochemical double layer capacitor, *Sustainable Energy Fuels* **2**, 968 (2018).
 - [7] P. Zaccagnini, D. di Giovanni, M. G. Gomez, S. Passerini, A. Varzi, and A. Lamberti, Flexible and high temperature supercapacitor based on laser-induced graphene electrodes and ionic liquid electrolyte, a de-rated voltage analysis, *Electrochim. Acta* **357**, 136838 (2020).
 - [8] S. Kondrat, G. Feng, F. Bresme, M. Urbakh, and A. A. Kornyshev, Theory and simulations of ionic liquids in nanoconfinement, *Chem. Rev.* **123**, 6668 (2023).
 - [9] L. L. Zhang and X. Zhao, Carbon-based materials as supercapacitor electrodes, *Chem. Soc. Rev.* **38**, 2520 (2009).
 - [10] C. Largeot, C. Portet, J. Chmiola, P.-L. Taberna, Y. Gogotsi, and P. Simon, Relation between the ion size and pore size for an electric double-layer capacitor, *J. Am. Chem. Soc.* **130**, 2730 (2008).
 - [11] P. Tamailarasan and S. Ramaprabhu, Carbon nanotubes-graphene-solidlike ionic liquid layer-based hybrid electrode material for high performance supercapacitor, *J. Phys. Chem. C* **116**, 14179 (2012).
 - [12] C. Liu, Z. Yu, D. Neff, A. Zhamu, and B. Z. Jang, Graphene-based supercapacitor with an ultrahigh energy density, *Nano Lett.* **10**, 4863 (2010).
 - [13] T. Pajkossy, C. Müller, and T. Jacob, The metal–ionic liquid interface as characterized by impedance spectroscopy and in

- situ scanning tunneling microscopy, *Phys. Chem. Chem. Phys.* **20**, 21241 (2018).
- [14] C. Liu, S. Qiu, P. Du, H. Zhao, and L. Wang, An ionic liquid–graphene oxide hybrid nanomaterial: Synthesis and anti-corrosive applications, *Nanoscale* **10**, 8115 (2018).
- [15] J. Ye, Y.-C. Wu, K. Xu, K. Ni, N. Shu, P.-L. Taberna, Y. Zhu, and P. Simon, Charge storage mechanisms of single-layer graphene in ionic liquid, *J. Am. Chem. Soc.* **141**, 16559 (2019).
- [16] M. Z. Bazant, B. D. Storey, and A. A. Kornyshev, Double layer in ionic liquids: Overscreening versus crowding, *Phys. Rev. Lett.* **106**, 046102 (2011).
- [17] A. A. Kornyshev, Double-layer in ionic liquids: Paradigm change? *J. Phys. Chem. B* **111**, 5545 (2007).
- [18] S. May, Differential capacitance of the electric double layer: mean-field modeling approaches, *Curr. Opin. Electrochem.* **13**, 125 (2019).
- [19] R. Downing, G. V. Bossa, and S. May, The role of ion–ion correlations for the differential capacitance of ionic liquids, *J. Phys. Chem. C* **122**, 28537 (2018).
- [20] Y. S. Park and I. S. Kang, Perturbation analysis for the effects of ion correlations on the surface force and the specific capacitance in a nanochannel, *Colloids Surf. A* **614**, 126207 (2021).
- [21] C. S. Perez-Martinez, A. M. Smith, and S. Perkin, Underscreening in concentrated electrolytes, *Faraday Discuss.* **199**, 239 (2017).
- [22] H. Zhao, Diffuse-charge dynamics of ionic liquids in electrochemical systems, *Phys. Rev. E* **84**, 051504 (2011).
- [23] A. A. Lee, S. Kondrat, D. Vella, and A. Goriely, Dynamics of ion transport in ionic liquids, *Phys. Rev. Lett.* **115**, 106101 (2015).
- [24] M. Yavarian, R. Melnik, and Z. L. Mišković, Frequency response of the diffuse-charge dynamics in electrochemical systems with a graphene electrode, *J. Electrochem. Soc.* **171**, 086501 (2024).
- [25] M. Yavarian, R. Melnik, and Z. L. Mišković, Modeling of charging dynamics in electrochemical systems with a graphene electrode, *J. Electroanal. Chem.* **946**, 117711 (2023).
- [26] J. P. de Souza and M. Z. Bazant, Continuum theory of electrostatic correlations at charged surfaces, *J. Phys. Chem. C* **124**, 11414 (2020).
- [27] R. F. Stout and A. S. Khair, A continuum approach to predicting electrophoretic mobility reversals, *J. Fluid Mech.* **752**, R1 (2014).
- [28] J.-L. Liu and B. Eisenberg, Correlated ions in a calcium channel model: a Poisson–Fermi theory, *J. Phys. Chem. B* **117**, 12051 (2013).
- [29] J.-L. Liu and B. Eisenberg, Numerical methods for a Poisson–Nernst–Planck–Fermi model of biological ion channels, *Phys. Rev. E* **92**, 012711 (2015).
- [30] J.-L. Liu and B. Eisenberg, Analytical models of calcium binding in a calcium channel, *J. Chem. Phys.* **141**, 075102 (2014).
- [31] J.-L. Liu and B. Eisenberg, Poisson–Nernst–Planck–Fermi theory for modeling biological ion channels, *J. Chem. Phys.* **141**, 22D532 (2014).
- [32] A. Shalabi, L. Daniels, M. Scott, and Z. Mišković, Differential capacitance of ionic liquid interface with graphene: The effects of correlation and finite size of ions, *Electrochim. Acta* **319**, 423 (2019).
- [33] Z. A. Goodwin and A. A. Kornyshev, Underscreening, overscreening and double-layer capacitance, *Electrochem. Commun.* **82**, 129 (2017).
- [34] M. A. Gebbie, A. M. Smith, H. A. Dobbs, G. G. Warr, X. Banquy, M. Valtiner, M. W. Rutland, J. N. Israelachvili, S. Perkin, and R. Atkin, Long range electrostatic forces in ionic liquids, *Chem. Commun.* **53**, 1214 (2017).
- [35] B. D. Storey and M. Z. Bazant, Effects of electrostatic correlations on electrokinetic phenomena, *Phys. Rev. E* **86**, 056303 (2012).
- [36] A. Gupta, A. Govind Rajan, E. A. Carter, and H. A. Stone, Thermodynamics of electrical double layers with electrostatic correlations, *J. Phys. Chem. C* **124**, 26830 (2020).
- [37] M. Z. Bazant, K. Thornton, and A. Ajdari, Diffuse-charge dynamics in electrochemical systems, *Phys. Rev. E* **70**, 021506 (2004).
- [38] S. Schlumpberger and M. Z. Bazant, Simple theory of ionic activity in concentrated electrolytes, *arXiv:1709.03106*.
- [39] We note that Eq. (49), with γ_1 and γ_2 defined in Eq. (40), is a continuous function of δ_c , but exhibits singularities at the values $\delta_c = 0$ and $\delta_c = 0.5$, which are analytically removable in the sense of the L'Hôpital's rule, but cannot be handled by our numerical code. Given that we verified that the potential in Eq. (49) is a smooth function of δ_c , we adopted the values 0.001 and 0.499 to represent the cases of $\delta_c = 0$ and $\delta_c = 0.5$, respectively. In this way, we avoid the singular points in Eq. (49) while giving accurate enough results.
- [40] A. Gupta and H. A. Stone, Electrical double layers: effects of asymmetry in electrolyte valence on steric effects, dielectric decrement, and ion–ion correlations, *Langmuir* **34**, 11971 (2018).
- [41] A. Gupta, A. Govind Rajan, E. A. Carter, and H. A. Stone, Ionic layering and overcharging in electrical double layers in a Poisson–Boltzmann model, *Phys. Rev. Lett.* **125**, 188004 (2020).